

INCREMENTAL STRUCTURE FROM MOTION: A THREE-IMAGE PIPELINE SCALED TO MULTI-VIEW RECONSTRUCTION

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ABSTRACT

We present an incremental Structure-from-Motion (SfM) pipeline built from first principles — SIFT detection, descriptor matching, essential-matrix verification, parallax-gated two-view initialisation, PnP registration, and triangulation — and benchmark it against COLMAP. On a 20-frame sequential set it registers all cameras at 0.55 px median reprojection error; on a 169-image walk-around it registers 78/169 cameras at 0.61 px after correcting a camera-intrinsics error that alone halved reprojection error. Two findings frame the gap to COLMAP. First, an ablation shows retriangulation *alone* yields no improvement (78→75 cameras) — direct first-principles evidence that global *bundle adjustment*, not retriangulation, is the missing ingredient. Second, on 20 identical cameras COLMAP builds $\sim 3.6\times$ our points purely because its detector finds $\sim 5\times$ more keypoints per image, so the density gap is feature extraction, not the reconstruction engine. We further show the 91 unregistered images are not correspondence-starved but *geometrically incompatible* (PnP inlier ratio median 0.00 vs. 0.55 for registered images, a clean bimodal split), placing the coverage gap beyond any gate tuning and motivating multi-model reconstruction.

1 PROJECT SYNTHESIS AND BACKGROUND

1.1 PROBLEM DEFINITION

Structure from Motion (SfM) is the problem of recovering a sparse 3D point cloud and the corresponding camera poses from a set of 2D images without using depth sensors. The scene structure and camera motion are estimated jointly by triangulating feature correspondences across multiple views whose relative geometry is itself recovered from those correspondences. For a monocular, uncalibrated camera setup, the reconstruction is determined only up to an unknown global scale factor and a global rigid transformation (rotation and translation). This gauge freedom is typically fixed by defining the first camera as the world reference frame, setting $(\mathbf{R} = \mathbf{I}, \mathbf{t} = \mathbf{0})$.

1.2 END-TO-END METHOD

Across the project we built the full pipeline as composable stages:

Feature Detection and Matching (Week 2). SIFT keypoints and descriptors (Lowe, 2004) are detected per image through OpenCV’s SIFT function, with the default cap of 4000 keypoints/image. The resulting descriptors are matched across image pairs with a nearest-neighbour search, filtered by Lowe’s ratio test (default 0.75) to reject ambiguous matches. This is followed by geometric verification with a fundamental-matrix RANSAC fit (Fischler & Bolles, 1981), which rejects matches inconsistent with the epipolar constraint and yields a verified inlier count per image pair. The resulting inlier counts are used to shortlist candidate pairs for seed selection.

Two-view reconstruction (Week 3). From a chosen seed pair we recover the relative pose by estimating the essential matrix $\mathbf{E} = \mathbf{K}^\top \mathbf{F} \mathbf{K}$ from the verified inlier matches (a pinhole-camera intrinsics matrix $\mathbf{K}$ is assumed known), then decomposing it into the relative rotation and

translation $[\mathbf{R} | \mathbf{t}]$ via `cv2.recoverPose`, which selects the cheirality-consistent solution. We triangulate the inlier matches into an initial sparse 3D point cloud and fix the gauge by placing the first seed camera at the origin ($\mathbf{R} = \mathbf{I}$, $\mathbf{t} = \mathbf{0}$) and the second at $(\mathbf{R}, \mathbf{t})$.

Incremental growth (Week 4). Remaining images are registered one at a time by 2D–3D PnP and used to triangulate new structure.

Outputs. Per-step metrics, an RGB point cloud (PLY), camera-pose frusta, and reprojection-error statistics for evaluation.

We adopt the world→camera convention throughout: a point $\mathbf{X}$ projects in image i as $\mathbf{x} \propto \mathbf{K}(\mathbf{R}_i \mathbf{X} + \mathbf{t}_i)$ with points as column vectors, so camera i ’s centre is $\mathbf{C}_i = -\mathbf{R}_i^\top \mathbf{t}_i$. A single shared intrinsics matrix $\mathbf{K}$ is assumed (one camera, one resolution).

2 WEEK 4: EXTENDING TO DATASET RECONSTRUCTION

Week 3 reconstructed exactly three images. Week 4 generalises this to an *incremental* reconstruction over a pool of images. The central data structure is a **persistent point map**: every 3D Track stores the keypoint observations that created it, and every `RegisteredImage` keeps a `kpidx_to_point` table answering “does this keypoint already correspond to a known 3D point?” in $O(1)$. This is what makes greedy next-image scoring cheap at scale, where Week 3’s loose index arrays would not.

2.1 SEED SELECTION

Ranking candidate pairs by RANSAC inlier count alone is a trap: near-coincident views match beautifully but have near-parallel rays and triangulate poorly. We therefore *gate on parallax*. For each shortlisted pair we recover its two-view geometry, triangulate, and compute the **median triangulation angle** between the rays to each camera centre — a scale-invariant measure (and COLMAP’s `init_min_triangle` criterion). `choose_initial_pair` keeps pairs clearing a 5° gate and, among those, selects the one with the most pose inliers, falling back to the best pair if none clear the gate so the caller is never left without a seed. On the 20-frame set the top pair by inliers has only 2.41° parallax and is correctly rejected; the chosen seed has 946 pose inliers and 10.54° parallax.

2.2 INCREMENTAL GROWTH

For each unregistered image u , `gather_2d3d` accumulates its 2D–3D correspondences by reading `point_ids` from `kpidx_to_point` across *all* registered images. The greedy policy registers the image with the most such correspondences. Registration runs `cv2.solvePnP`; on success we link each PnP inlier into the point map. We then grow structure with `triangulate_new_points`, building both projection matrices explicitly as $\mathbf{P} = \mathbf{K}[\mathbf{R} | \mathbf{t}]$ (Week 3’s triangulator hardcodes the first camera at the origin and is wrong between two already-placed cameras). New points are kept only if finite, of positive depth in both cameras (cheirality), within the reprojection threshold, and — crucially — above a per-point triangulation-angle floor: reprojection and cheirality *do not* reject low-parallax points, which slide along the ray at near-constant reprojection and settle at garbage depth. A separate `extend_tracks` step links u ’s remaining keypoints to 3D points already owned by its neighbours (reprojection-gated), lengthening tracks and densifying the frontier so the next image is more likely to register.

2.3 REJECTION HANDLING

An image too weak to register now may succeed once more structure exists. The loop uses **reject-and-retry**: a failed image is added to a `rejected` set (so it cannot be re-picked until the model changes, preventing infinite spinning) and the set is *cleared whenever any registration succeeds*. The original design instead issued a hard `break` the instant the best candidate fell below `min_corr`, which abandoned every remaining image at once (§3 quantifies the damage); the current loop returns “stop” only when *no* candidate reaches the floor.

Table 1: Pipeline results. Reprojection error is per-observation, in pixels.

Dataset	Images	Registered	Sparse points	Median (px)	Mean (px)
partial_doge	20	20	7 254	0.553	1.039
Doge (full)	169	78	24 587	0.611	1.031

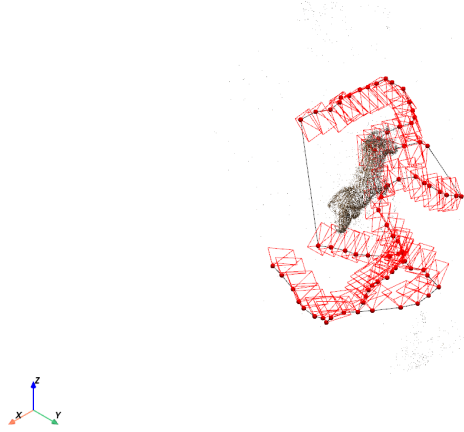


Figure 1: Final reconstruction on the full Doge set: 78 registered camera frusta (red), the registration-order trajectory (black), and the sparse cloud (grey, the statue recognisable at centre). The trajectory’s long leaps mark the capture gaps; the four unregistered blocks appear as the empty arcs of the walk-around.

3 RESULTS AND SYSTEM ANALYSIS

3.1 RECONSTRUCTED RESULTS

Table 1 summarises both reconstructions.

On the 20-frame sequential set the pipeline registers all 20 cameras, and the greedy order recovers the capture (timestamp) order because consecutive video frames share the most structure. The median/mean gap (0.55 vs 1.04 px) is a tail of higher-error points — the absence of joint refinement (see §5).

3.2 SCALING FAILURES AND FIXES

Pointing the working 20-frame pipeline at the full 169-image set exposed three distinct failures, each diagnosed from a measured symptom.

The stall (coverage). The run first stopped at 87/169 with whole contiguous capture-time blocks empty. This was *not* drift (the 2D–3D count is combinatorial and cannot fall with pose error) and *not* a disconnected graph (17 unregistered images had ≥ 100 verified inliers to a registered image). The cause was the hard break above. Replacing it with a soft stop, lowering the gates, and adding `_extend_tracks` were **necessary but not sufficient**: coverage improved but the run still stalled ($\sim 76/169$), because the remaining blocks are real viewpoint discontinuities, not loop behaviour.

Intrinsics (the largest quality fix). Registered cameras were initially low quality (PnP inlier ratios of a few percent). `K` had been built with a hardcoded principal point (800, 600) and a focal assuming

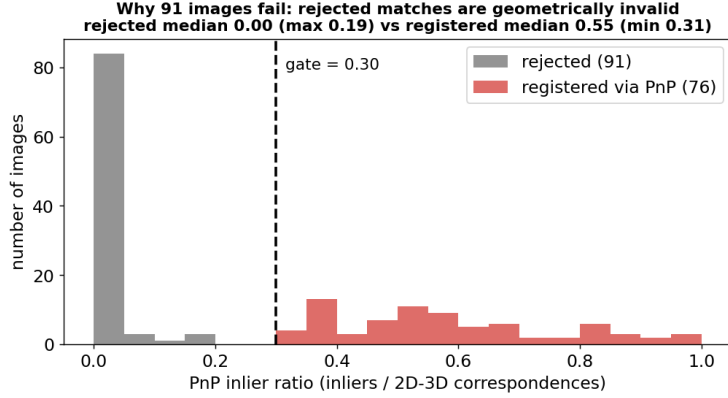


Figure 2: PnP inlier ratio for registered (red) vs. rejected (grey) images on the full set. The distributions are bimodal with no overlap; the 0.30 gate sits in the empty gap. Rejected images are geometrically incompatible, not correspondence-starved.

a 4:3, 2000-px-diagonal frame, whereas the Doge photos are 16:9 (native 4000×2256 , resized to 1600×902); c_y was off by ~ 149 px ($\approx 16\%$ of frame height), feeding a systematically wrong camera into every PnP and triangulation. Deriving the focal from a 24 mm-equivalent ($\rightarrow 1018.8$ px) and auto-centring the principal point to $(799.5, 450.5)$ moved the median reprojection error from **1.273 to 0.611 px** and made registrations uniformly clean. We now persist **K** in `cameras.json` so intrinsics are never again a forensic guess.

PnP inlier-ratio gate (quality). An absolute inlier floor alone accepts drifted poses that clear the floor (e.g. 37 inliers) while explaining only 3–5% of their correspondences. A second gate rejecting inliers $< 0.30 \times \text{count}$ catches the opposite failure to the floor. Together they moved the PnP inlier-ratio minimum from 0.03 to 0.31 and eliminated all sub-0.30 registrations (vs. 53/98 before).

3.3 FAILURE CASES: WHY 91 IMAGES NEVER REGISTER

The 91 unregistered images form four contiguous capture-gap blocks (e.g. a 3.3 s jump where the camera was repositioned). The natural hypothesis — that they are starved of correspondences — is *false*: probing each against the final cloud, the rejected images have a median of ~ 376 2D–3D footholds, comfortably above the `min_corr` floor. What fails is the geometry. Their PnP inlier ratio is median **0.00** (max 0.19), whereas every registered image sits at ≥ 0.31 (median 0.55). The two populations are cleanly *bimodal*, and the 0.30 ratio gate falls in the empty $[0.19, 0.31]$ band between them (Fig. 2). The matches exist but are geometrically incompatible with the reconstructed pose, so no threshold setting can admit them without admitting garbage — the gate is robust, not arbitrarily placed. The fix is structural: a separate model seeded on the rejected images registers a further +29, confirming they form a coherent but disjoint cluster (§5).

A dead-end worth recording: loosening the Lowe ratio to 0.85 made coverage *worse* (18/169), because PnP does not re-verify matches with a fundamental matrix the way the seed step does, so false correspondences poison it directly; and raising the feature cap to 8000 was bit-identical, because SIFT finds ≤ 3363 keypoints/image here, so the 4000 cap never bound. The weak bridges reflect genuine low visual overlap across the capture gaps, not a tunable matcher limit.

4 COMPARISON WITH COLMAP

We ran COLMAP (Schönberger & Frahm, 2016) (exhaustive matching, default pipeline) on the same images as each of our two reconstruction attempts. On the full 169-image set COLMAP registers **all 169** in a single model with 114 220 sparse points in 5.3 min, versus our 78/169 and 24 587 points (Table 2). Its cloud is $\sim 4.6\times$ denser and it crosses the capture gaps our pipeline cannot.

Table 2: Our pipeline vs. COLMAP on the full Doge set (169 images).

	Registered	Sparse points	Median reproj. (px)	Wall-clock
Ours	78/169	24 587	0.611	~5 min
COLMAP	169/169	114 220	~1.0	5.3 min

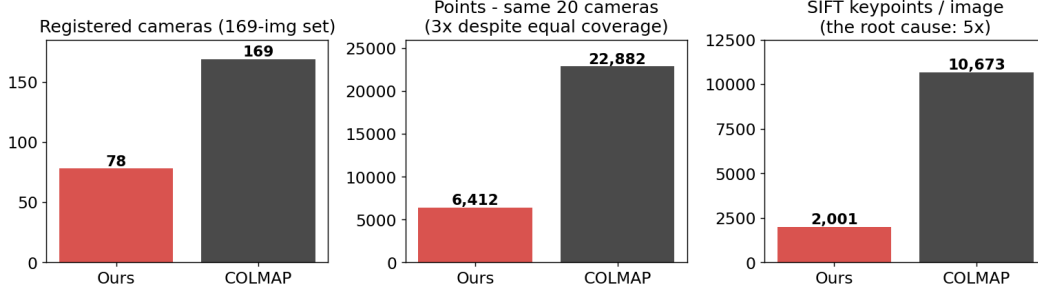
COLMAP vs ours: density gap is feature detection, not the SfM engine

Figure 3: COLMAP vs. ours. *Left*: registered cameras on the full set (coverage gap). *Centre*: points on the same 20 cameras — a density gap that persists despite identical coverage. *Right*: its root cause — COLMAP’s detector finds $\sim 5\times$ more keypoints per image.

The density gap is feature detection, not the engine. The full-set comparison confounds two effects — coverage and density — so we isolate density on the 20-frame set, where *both* systems register all 20 cameras. Even with identical coverage COLMAP builds 22 882 points to our 6 412 (Fig. 3, $\sim 3.6\times$). The cause is upstream: COLMAP’s detector finds $\sim 10\,673$ keypoints/image against our SIFT’s $\sim 2\,001$ ($\sim 5\times$), and once the keypoint budget is accounted for our triangulation efficiency (kept points per keypoint) is comparable. The density gap is therefore a *feature-extraction* gap, fixable by lowering the SIFT contrast threshold, *not* a limitation of our reconstruction engine.

The *coverage* difference has a specific mechanism, not merely “COLMAP is better”: COLMAP interleaves global bundle adjustment with retriangulation, repeatedly refining all poses and points and then re-triangulating against the improved poses, densifying the cloud until previously gap-isolated frames accumulate enough valid structure to register. The load-bearing component is the *global optimisation* — our own ablation (§5) shows retriangulation without it does nothing. Our single greedy forward pass has no joint-refinement step, so a true low-overlap gap is terminal for us but bridgeable for COLMAP.

5 DISCUSSION

5.1 WHAT IS MISSING TO REACH COLMAP-QUALITY

Bundle adjustment is the missing piece — and we have direct evidence. We implemented re-triangulation (re-triangulating existing tracks against the current cloud) and ablated it. It made *no* difference: 78 vs. 75 cameras and 24 587 vs. 23 420 points, within run-to-run noise (Fig. 4). This null result is informative rather than disappointing: re-triangulating against poses we never refined cannot improve them, so the gain COLMAP extracts from retriangulation is really a gain from the *global bundle adjustment* it interleaves. We never jointly refine poses and points, so small per-step errors accumulate as drift — the visible median (0.55 px) vs. mean (1.04 px) reprojection gap — and retriangulation has no refined geometry to exploit. **Global BA is therefore the single largest missing piece**; only once it exists does retriangulation become worthwhile.

Two further capabilities.

Multi-model reconstruction with Sim(3) merging. The coverage gap is a disjoint-cluster problem (Fig. 2): re-seed a fresh model from the best unregistered pair, grow it (a sec-

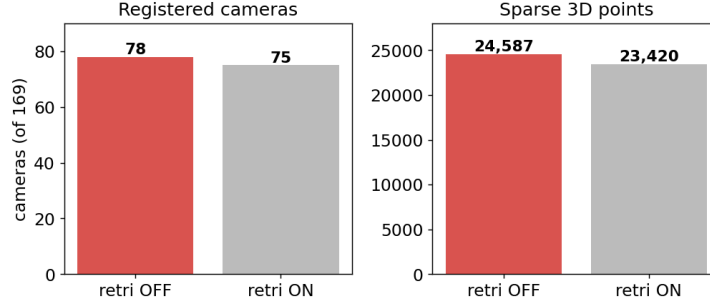
Retriangulation alone $\approx$ no change — it needs bundle adjustment

Figure 4: Retriangulation ablation on the full set. With no bundle adjustment to refine the poses it triangulates against, retriangulation produces no improvement in either cameras or points — evidence that BA, not retriangulation, is the load-bearing addition.

ond model recovers +29 images here), and align models with a 7-DoF similarity (Umeyama (Umeyama, 1991) + RANSAC) before a global BA closes the seam. This is the direct fix for the capture-gap blocks a single incremental model cannot cross.

Denser features. As the COLMAP comparison showed, the density gap is upstream of the engine; a lower SIFT contrast threshold (or a learned detector) closes most of it.

We also lack track merging across loops (a point re-seen after the camera loops back is duplicated, not merged) and per-image intrinsics (a single shared $\mathbf{K}$ assumes one camera/resolution).

5.2 FIRST-PRINCIPLES REFLECTION

Our Week 1 study showed COLMAP itself failing on textureless surfaces, repetitive patterns, low-overlap captures, and near-pure rotation. These are not COLMAP bugs but consequences of the two hand-crafted primitives it shares with our pipeline. *SIFT* keys on local gradient structure, so a textureless or specular surface yields no stable keypoints and a repetitive pattern yields many mutually ambiguous ones; *RANSAC on an epipolar/PnP model* can verify only what the matcher proposes and degenerates when the motion is near-pure-rotation (no parallax, essential matrix ill-conditioned) or the overlap is too low to outvote false matches. Our pipeline exhibits sharper versions of the same modes: loosening the Lowe ratio to 0.85 collapsed coverage to 18/169 (the false-match failure, since PnP does not re-verify with a fundamental matrix), and our capture-gap blocks are exactly the low-overlap case — bimodal-bad PnP ratios (Fig. 2) where the proposed matches are geometrically incompatible.

From first principles, then, two layers separate a robust system from where we are. The first is the gap *to* COLMAP: a purely incremental, locally-greedy estimator has no global consistency constraint, so errors are neither bounded (drift) nor repaired (gaps, duplicates); global bundle adjustment and multi-model merging close this by optimising structure and motion jointly rather than committing irreversibly to each local decision. The second is the gap *past* COLMAP, and it targets the Week 1 failures directly: because they originate in the correspondence front-end, the principled remedy is *learned* correspondence — detectors and matchers (e.g. SuperPoint (DeTone et al., 2018), SuperGlue (Sarlin et al., 2020), LoFTR (Sun et al., 2021)) that use semantic context to find repeatable features on weak texture and to disambiguate repeated structure — plus motion priors and loop closure to constrain the degenerate-motion and revisited-scene cases that no purely local, hand-crafted estimator can resolve.

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A ITERATION LOG: SCALING FROM 20 TO 169 IMAGES

Each fix below was driven by a measured symptom, not a guess. Coverage is *cameras registered / 169*; quality is *median per-observation reprojection error*.

Table 3: Symptom $\rightarrow$ diagnosis $\rightarrow$ change $\rightarrow$ evidence for the full-set scaling.

Symptom	Diagnosis	Change	Evidence
Stall at 87/169, whole capture-time blocks empty	Not drift (2D–3D count is combinatorial) and not a disconnected graph (17 unreg. images had ≥ 100 inliers to a reg. image); a <code>hard break</code> abandoned all remaining images at the first sub-floor pick	Soft stop (reject-and-retry, cleared on any success); lower gates; add <code>_extend_tracks</code> to densify the frontier	Necessary but not sufficient: $\sim 76/169$ — remaining blocks are real capture gaps
Even registered cameras low quality (PnP ratios a few %)	K hardcoded for 4:3/2000px; data is 16:9, c_y off by ~ 149 px ($\approx 16\%$ of height)	Derive focal from 24 mm-equiv $\rightarrow 1018.8$ px; auto-centre principal point; persist K in <code>cameras.json</code>	Median reproj. 1.273 $\rightarrow$ 0.611 px; registrations uniformly clean
Drifted poses cleared the inlier floor (e.g. 37 inliers, 3–5% ratio)	Absolute floor cannot reject many-point/low-fraction poses	Add PnP inlier-ratio gate ($< 0.30 \times \text{count}$)	Ratio min 0.03 $\rightarrow$ 0.31, median 0.27 $\rightarrow$ 0.56; zero sub-0.30 (vs 53/98)
Low-parallax points at garbage depth survive reproj. + cheirality	Near-parallel rays slide along the ray at near-constant reprojection	Per-point triangulation-angle floor ($\geq 2^\circ$) in <code>triangulate_new_points</code>	Cleaner cloud at small coverage cost
Weak bridges to stalled blocks (≤ 19 inliers)	Hypothesis: more/looser matching lifts them	Lowe 0.75 $\rightarrow$ 0.85; features 4000 $\rightarrow$ 8000	Worse (18/169) / bit-identical (SIFT finds ≤ 3363 kp): dead end

Re-validation (correct **K + final gates).** 78/169 cameras, 24 587 points, median 0.611 px / mean 1.031 px; every registration clean (PnP ratio min 0.31). The 91 unregistered images remain four contiguous capture-gap blocks (151332–151413, 151432–151453, the 151517 two-frame island, 151549–151552), confirming bad-**K** hurt *quality* while the *coverage* gap is independently the capture gaps.

B GROUP DYNAMICS